

# COMBINATORIAL CATEGORICAL EQUIVALENCES

STEPHEN LACK AND ROSS STREET

## CONTENTS

|                                |   |
|--------------------------------|---|
| 1. Introduction                | 1 |
| 2. A setting for one direction | 1 |
| 3. A tighter setting           | 3 |
| References                     | 5 |

## 1. INTRODUCTION

The purpose is to generalize our work in [15] to include the Dold-Puppe-Kan Theorem [5, 6, 14] on additive simplicial objects and chain complexes.

## 2. A SETTING FOR ONE DIRECTION

Let  $\mathcal{P}$  be a category. Let  $\mathcal{Z}$  be a class of non-invertible retractions in  $\mathcal{P}$ , closed under composition. Write  $\bar{\mathcal{Z}}$  for the left ideal generated by  $\mathcal{Z}$ . That is,  $\bar{\mathcal{Z}}$  consists of all the morphisms  $g \circ z$  with  $z \in \mathcal{Z}$ . Let  $\mathcal{R}$  be a class of morphisms  $r$  in  $\mathcal{P}$  having the following properties:

- (a)  $\mathcal{R} \cap \bar{\mathcal{Z}} = \emptyset$ ;
- (b) If  $r : X \rightarrow A$  is in  $\mathcal{R}$  and  $z : A \rightarrow U$  is in  $\mathcal{Z}$  then  $z \circ r : X \rightarrow U$  is in  $\bar{\mathcal{Z}}$ .

**Example 2.1.** As in [15], take a category  $\mathcal{A}$  with a factorization system  $(\mathcal{E}, \mathcal{M})$ . Assume that the pullback of any morphism with one in  $\mathcal{M}$  exist. Assume every morphism in  $\mathcal{M}$  is a monomorphism. Let  $\mathcal{P}$  be the category  $\text{Par}_{\mathcal{M}}\mathcal{A}$  of  $\mathcal{M}$ -partial maps in  $\mathcal{A}$ . Take  $\mathcal{Z}$  to consist of the partial maps  $m^* : A \rightarrow U$  for all non-invertible  $m : U \rightarrow A$  in  $\mathcal{M}$ . Let  $\mathcal{R}$  consist of the partial maps  $e_* : X \rightarrow Y$  for  $e : X \rightarrow Y$  in  $\mathcal{E}$ .

To see that property (a) holds, assume  $e_* = g \circ m^*$  with  $e \in \mathcal{E}$  and  $m \in \mathcal{M}$ . Then we have the diagram

$$\begin{array}{ccccc}
 & & A & & \\
 & \swarrow^{1_A} & & \searrow_e & \\
 & U & & V & \\
 & \swarrow_m & \searrow_{1_U} & \swarrow_{g_0} & \searrow_{g_1} \\
 A & & U & & B
 \end{array}
 \tag{2.1}$$

---

*Date:* November 18, 2013.

Both authors gratefully acknowledge the support of the Australian Research Council Discovery Grant DP130101969; Lack acknowledges with equal gratitude the support of an Australian Research Council Future Fellowship.

in which the diamond is a pullback. So  $p_1$  is invertible and  $mp_0 = 1_A$ . Since  $m$  is a monomorphism and a retraction, it is invertible. So  $m^* \notin \mathcal{Z}$ .

To see that the property (b) holds, consider  $r = e_*$  and  $z = m^*$  with  $e \in \mathcal{E}$  and  $m \in \mathcal{M}$ . We have

$$m^* \circ e_* = q_* \circ n^*$$

as per the pullback

$$\begin{array}{ccc} P & \xrightarrow{q} & U \\ n \downarrow & & \downarrow m \\ X & \xrightarrow{e} & A. \end{array} \quad (2.2)$$

If  $n$  is invertible then  $mq = en \in \mathcal{E}$ , so  $m \in \mathcal{E} \cap \mathcal{M}$  is invertible. So  $m^* \in \mathcal{Z}$  implies  $n^* \in \mathcal{Z}$ .

**Example 2.2.** Take  $\mathcal{P}$  to be the category  $\Delta_{\perp, \top}$  of finite non-empty ordinals  $n = \{0, 1, \dots, n-1\}$  with morphisms those functions which preserve first element, last element and order. Take  $\mathcal{Z}$  to consist of the non-invertible surjections  $\sigma : m \rightarrow n$  such that  $\sigma(i) = 0$  implies  $i = 0$ ; that is,  $\sigma(1) = 1$ . In particular, for  $k \neq 0$ , the surjections  $\sigma_k : m+1 \rightarrow m$  with  $\sigma_k(k) = \sigma_k(k+1)$  are in  $\mathcal{Z}$ ; and every member of  $\mathcal{Z}$  is a composite of these. Take  $\mathcal{R}$  to consist of the isomorphisms and the surjections  $\sigma_0 : m+1 \rightarrow m$  such that  $\sigma_0(1) = 0$ .

To see that property (a) holds, suppose  $\sigma_0 = \xi\sigma$  with  $\sigma : m+1 \rightarrow n$  a non-invertible surjection. The last statement implies  $n < m+1$ . Since  $\sigma_0$  is surjective,  $\xi$  is surjective. So  $n \geq m$ . So  $m = n$  and  $\xi$  is an identity. This means  $\sigma = \sigma_0$  is not in  $\mathcal{Z}$ .

To see that property (b) holds, just notice that, for  $0 < k$ ,

$$\sigma_k \sigma_0 = \sigma_0 \sigma_{k+1}.$$

Returning to the general setting, we define a pointed category (that is, a category enriched in pointed sets)  $\mathcal{D}$  with the same objects as  $\mathcal{P}$  and the non-zero morphisms those of  $\mathcal{R}$ . Composition is as in  $\mathcal{P}$  when the result is in  $\mathcal{R}$  but it is zero otherwise.

Let  $\mathcal{X}$  be an additive category with sufficient limits including finite direct sums. There is a functor

$$\widetilde{(-)} : [\mathcal{P}, \mathcal{X}] \rightarrow [\mathcal{D}, \mathcal{X}]_{\text{pt}} \quad (2.3)$$

where the codomain category consists of the pointed functors. The definition on objects is:

$$\widetilde{T}A = \bigcap_{(A \xrightarrow{z} U) \in \mathcal{Z}} \ker(Tz) \quad (2.4)$$

and, for  $r : A \rightarrow B$  in  $\mathcal{R}$ , the morphism  $\widetilde{T}r : \widetilde{T}A \rightarrow \widetilde{T}B$  is the restriction of  $Tr : TA \rightarrow TB$ . Why does it restrict? Let  $i_A : \widetilde{T}A \rightarrow TA$  be the inclusion. Take any  $z : B \rightarrow V$  in  $\mathcal{Z}$ . Then  $(Tz)(Tr)i_A = T(zr)i_A = T(g\bar{z})i_A = (Tg)(T\bar{z})i_A = 0$  since  $\bar{z} \in \mathcal{Z}$  using property (b). So there exists  $\widetilde{T}r$  such that  $i_B(\widetilde{T}r) = (Tr)i_A$ , as claimed.

## 3. A TIGHTER SETTING

We again start with a category  $\mathcal{P}$ . We assume we have two classes  $\mathcal{M}$  and  $\mathcal{R}$  of morphisms in  $\mathcal{P}$ .

We ask that  $\mathcal{M}$  be closed under composition, contain the isomorphisms, and consist of coretractions (= split monomorphisms).

We ask that  $\mathcal{R}$  should contain the isomorphisms, contain composites on both sides with isomorphisms, and have the property that  $g \circ f \in \mathcal{R}$  implies  $g \in \mathcal{R}$ .

This time we *define*  $\mathcal{Z}$  to consist of those morphisms  $z : X \rightarrow U$  which have a right inverse  $z_* \in \mathcal{M}$  and, if  $g \circ z = f \circ m$  with  $m \in \mathcal{M}$ , then  $f \notin \mathcal{R}$ . Clearly  $\mathcal{Z}$  includes no invertible morphisms yet is closed under composition. In fact, if  $g$  has a right inverse and  $z \in \mathcal{Z}$  then  $g \circ z \in \mathcal{Z}$ .

We require a set  $\mathcal{I}$  of pairs of morphisms  $(m, m^*)$  where  $m \in \mathcal{M}$ ,  $m^* \circ m = 1$ , and either  $m^* \in \mathcal{Z}$  or  $m^* = 1$ . We assume every  $z \in \mathcal{Z}$  has the form  $z = g \circ m^*$  for some retraction  $g$  and  $(m, m^*) \in \mathcal{I}$  with  $m^* \neq 1$ .

The extra property we require is

$$\begin{aligned} &\text{every } f \in \mathcal{P} \text{ factors uniquely as } f = m \circ r \circ z \text{ with } m \in \mathcal{M}, r \in \mathcal{R}, \\ &\text{and either } z \in \mathcal{Z} \text{ or } z \text{ an identity.} \end{aligned} \tag{3.5}$$

Property (a) of Section 2 is now a tautology. We also have property (b) of Section 2:

**Proposition 3.1.** *Given  $z : X \rightarrow U$  in  $\mathcal{Z}$  and  $r : A \rightarrow X$  in  $\mathcal{R}$ , there exist  $\zeta : A \rightarrow B$  in  $\mathcal{Z}$  and some  $g : B \rightarrow U$  in  $\mathcal{P}$  such that  $z \circ r = g \circ \zeta$ .*

*Proof.* Apply (3.5) to  $z \circ r$  to obtain  $z \circ r = m \circ s \circ \zeta$  with  $m \in \mathcal{M}$ ,  $s \in \mathcal{R}$ , and either  $\zeta \in \mathcal{Z}$  or  $\zeta$  an identity. If  $\zeta$  is an identity then  $m^* \circ z \circ r = s \in \mathcal{R}$  where  $m^*$  is a left inverse for  $m$ . It follows that  $m^* \circ z \in \mathcal{R}$ . But this contradicts  $z \in \mathcal{Z}$ . So  $\zeta$  is not an identity. Hence  $\zeta \in \mathcal{Z}$  and we take  $g = m \circ s$ .  $\square$

**Proposition 3.2.** *Given  $n : U \rightarrow X$  in  $\mathcal{M}$  and  $s : X \rightarrow Y$  in  $\mathcal{R}$ , there exist unique  $r : U \rightarrow V$  in  $\mathcal{R}$  and  $m : V \rightarrow Y$  in  $\mathcal{M}$  such that  $s \circ n = m \circ r$ .*

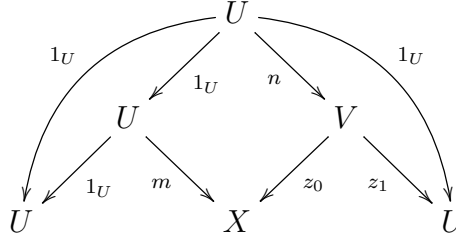
*Proof.* Apply (3.5) to  $s \circ n$  to obtain  $s \circ n = m \circ r \circ z$  with  $m \in \mathcal{M}$ ,  $r \in \mathcal{R}$ , and either  $z \in \mathcal{Z}$  or  $z$  an identity. By definition of  $\mathcal{Z}$  we cannot have both  $z \in \mathcal{Z}$  and  $s \in \mathcal{R}$ . So  $z$  is an identity. So take  $s \circ n = m \circ r$  as required.  $\square$

It behooves us to show that our two examples of Section 2 are still included.

**Example 3.3.** As in Example 2.1, take a category  $\mathcal{A}$  with a factorization system  $(\mathcal{E}, \mathcal{M})$ . Assume that the pullback of any morphism with one in  $\mathcal{M}$  exist. Assume every morphism in  $\mathcal{M}$  is a monomorphism. Let  $\mathcal{P}$  be the category  $\text{Par}_{\mathcal{M}} \mathcal{A}$  of  $\mathcal{M}$ -partial maps in  $\mathcal{A}$ . Identify  $\mathcal{M}$  in  $\mathcal{A}$  with the morphisms  $m_*$  in  $\mathcal{P}$  for  $m \in \mathcal{M}$ . Take  $\mathcal{R}$  to consist of the morphisms  $e_*$  in  $\mathcal{P}$  with  $e \in \mathcal{E}$ . With  $\mathcal{Z}$  as defined in this Section, for this example we claim

$$\mathcal{Z} = \{m^* : m \text{ non-invertible in } \mathcal{M}\}.$$

Take  $z : X \longrightarrow U$  in  $\mathcal{Z}$ . In order for a partial map  $z = [z_0, V, z_1]$  to have a right inverse in  $\mathcal{M}$ , we see from the diagram



with  $m, n \in \mathcal{M}$ , we must have  $n$ , and hence  $z_1$ , invertible. So  $z = m^*$ . So each  $z \in \mathcal{Z}$  is in the set claimed.

Now suppose we have  $e_* \circ m_* = g \circ n^*$  with  $e \in \mathcal{E}$  and  $m, n \in \mathcal{M}$ . It follows that  $n \circ g_0 = 1$ . So  $n^* \notin \mathcal{Z}$ . Our characterization of  $\mathcal{Z}$  follows.

We take  $\mathcal{I} = \{(m_*, m^*) : m \in \mathcal{M}\}$ .

Finally, every  $f$  in  $\mathcal{P}$  uniquely factors as  $f = m_* \circ e_* \circ n^*$  with  $m, n \in \mathcal{M}$  and  $e \in \mathcal{E}$ . But then  $n^*$  is either invertible or in  $\mathcal{Z}$ .

**Example 3.4.** As in Example 2.2, take  $\mathcal{P}$  to be the category  $\Delta_{\perp, \top}$  of finite non-empty ordinals  $n = \{0, 1, \dots, n-1\}$  with morphisms those functions which preserve first element, last element and order. We write  $\sigma_k : m+1 \rightarrow m$  for the order-preserving surjection which takes the value  $k$  twice. We write  $\partial_k : m \rightarrow m+1$  for the order-preserving injection which does not have  $k$  in its image. Take  $\mathcal{M}$  to consist of all the injective functions in  $\Delta_{\perp, \top}$ . Take  $\mathcal{R}$  to consist of the isomorphisms and the surjections  $\sigma_0 : m+1 \rightarrow m$ . With  $\mathcal{Z}$  as defined in this Section, for this example we claim

$$\mathcal{Z} = \{\tau \sigma_k : k > 0 \text{ and } \tau \text{ surjective}\}.$$

If  $z$  is to have a right inverse it must be a surjection and not be  $\sigma_0 : 2 \rightarrow 1$ . In fact no  $\sigma_0 : m+1 \rightarrow m$  can be in  $\mathcal{Z}$  since  $\partial_1 \sigma_0 = \sigma_0 \partial_2$  with  $\sigma_0 \in \mathcal{R}$  and  $\partial_2 \in \mathcal{M}$ .

Now take any non-invertible surjection  $\sigma$  with  $\sigma(1) = 1$ . We shall show that  $\sigma \in \mathcal{Z}$ . Assume  $\xi \sigma = \rho \partial$  where  $\rho \in \mathcal{R}$  and  $\partial \in \mathcal{M}$ . Clearly  $\rho$  cannot be invertible or else  $\sigma$  would be injective and so be invertible. So  $\rho = \sigma_0$ . Using the fact that  $\sigma \partial_i$  is an identity for  $i = 1$  and equal to  $\partial_{i-1} \sigma_0$  for  $i > 1$ , we see that  $\sigma_0 \partial$  is either injective or of the form  $\nu \sigma_0$  for some injective  $\nu$ . Factor  $\xi = \mu \tau$  with  $\tau$  surjective and  $\mu$  injective. We have  $\mu \tau \sigma$  either injective or equal to  $\nu \sigma_0$ . The first case is impossible since  $\sigma$  is not injective. By uniqueness of factorization, the second case implies  $\tau \sigma = \sigma_0$  and  $\mu = \nu$ . So  $\tau = 1$  and  $\sigma = \sigma_0$ , contradicting  $\sigma(1) = 1$ . So our assumption was impossible and we are left with  $\sigma \in \mathcal{Z}$ . In particular, this implies  $\sigma_k \in \mathcal{Z}$  for  $k > 0$  and hence  $\tau \sigma_k \in \mathcal{Z}$  for any surjective  $\tau$ .

Now suppose  $(\sigma : m+1 \rightarrow m) \in \mathcal{Z}$ . Let  $0 \leq k < m$  be the largest such that  $\sigma(k) = \sigma(k+1)$ . Then  $\sigma = \tau \sigma_k$  for some surjection  $\tau$ . If  $k = 0$  then  $\sigma = \sigma_0$  which we have seen is contrary to  $\sigma \in \mathcal{Z}$ . So  $\sigma = \tau \sigma_k$  with  $k > 0$ . This proves our characterization of  $\mathcal{Z}$ .

We take  $\mathcal{I} = \{(\partial_k, \sigma_k) : k > 0\}$ .

Finally, for the factorization, take any  $\xi$  in  $\Delta_{\perp, \top}$ . If  $\xi$  is injective, the required factorization is clear. If  $\xi = \partial \sigma_k$  with  $\partial$  injective, the factorization is clear, breaking into two cases depending on whether  $k = 0$  or  $k > 0$ . Otherwise we can uniquely factorize  $\xi = \partial \sigma_i \tau \sigma_k$  where  $\partial$  is injective,  $i < k$  and  $\tau$  is a surjection for which

$\tau(j) = \tau(j+1)$  implies  $i < j < k$ ; again we have the desired factorization in each of the two cases  $i = 0$  or  $i > 0$ .

Returning to the general situation of this Section, notice that the formula (2.4) for  $\tilde{T} : \mathcal{D} \longrightarrow \mathcal{X}$  now can be rewritten

$$\tilde{T}A = \bigcap_{(U \xrightarrow{m} A, m^*) \in \mathcal{I}, m^* \neq 1} \ker(Tm^*) . \quad (3.6)$$

We can also construct a functor

$$\widehat{(-)} : [\mathcal{D}, \mathcal{X}]_{\text{pt}} \longrightarrow [\mathcal{P}, \mathcal{X}] \quad (3.7)$$

whose value at the pointed functor  $F : \mathcal{D} \longrightarrow \mathcal{X}$  is the functor  $\hat{F} : \mathcal{P} \longrightarrow \mathcal{X}$  defined as follows. On objects, put

$$\hat{F}A = \sum_{(U \xrightarrow{m} A, m^*) \in \mathcal{I}} FU .$$

For a morphism  $f : A \longrightarrow B$  in  $\mathcal{P}$ , take the factorization  $f = n_1 r z$  with  $z : X \rightarrow U$  in  $\mathcal{X}$  or an identity,  $r \in \mathcal{R}$  and  $n_1 : V \rightarrow Y$  in  $\mathcal{M}$ . The morphism  $\hat{F}f : \hat{F}A \longrightarrow \hat{F}B$  has composite with the injection  $\text{in}_m : FU \rightarrow \hat{F}A$  equal to zero unless

## REFERENCES

- [1] Clemens Berger, *Opérades cellulaires et espaces de lacets itérés*, Annales de L’Institut Fourier **46(4)** (1996) 1125–1157.
- [2] Thomas Church, Jordan S. Ellenberg and Benson Farb, *FI-modules: a new approach to stability for  $S_n$ -representations*, ([arXiv:1204.4533v2](#), 28 Jun 2012).
- [3] Brian J. Day and Ross Street, *Kan extensions along promonoidal functors*, Theory and Applications of Categories **1(4)** (1995) 72–77.
- [4] Brian J. Day and Ross Street, *Monoidal bicategories and Hopf algebroids*, Advances in Math. **129** (1997) 99–157.
- [5] Albrecht Dold, *Homology of symmetric products and other functors of complexes*, Annals of Math. **68** (1958) 54–80. [1](#)
- [6] Albrecht Dold and Dieter Puppe, *Homologie nicht-additiver Funktoren. Anwendungen*, Ann. Inst. Fourier Grenoble **11** (1961) 201–312. [1](#)
- [7] Peter J. Freyd and G. Max Kelly, *Categories of continuous functors I*, J. Pure and Applied Algebra **2** (1972) 169–191; Erratum Ibid. **4** (1974) 121.
- [8] Christopher J. Hillar and Darren L. Rhea, *Automorphisms of finite abelian groups*, American Mathematical Monthly **114(10)** (2007) 917–923; [arXiv:math/0605185](#).
- [9] André Joyal, *Une théorie combinatoire des séries formelles*, Advances in Mathematics **42** (1981) 1–82.
- [10] André Joyal, *Foncteurs analytiques et espèces de structures*, Lecture Notes in Mathematics **1234** (Springer 1986) 126–159.
- [11] André Joyal and Ross Street, *Tortile Yang-Baxter operators in tensor categories*, J. Pure Appl. Algebra **71** (1991) 43–51.
- [12] André Joyal and Ross Street, *An introduction to Tannaka duality and quantum groups*; Part II of Category Theory, Proceedings, Como 1990 (Editors A. Carboni, M.C. Pedicchio and G. Rosolini) Lecture Notes in Math. **1488** (Springer-Verlag Berlin, Heidelberg 1991) 411–492.
- [13] André Joyal and Ross Street, *The category of representations of the general linear groups over a finite field*, J. Algebra **176 (3)** (1995) 908–946.
- [14] Daniel Kan, *Functors involving c.s.s complexes*, Transactions of the American Mathematical Society **87** (1958) 330–346. [1](#)

- [15] Stephen Lack and Ross Street, *Morita equivalence of partial maps and strong epimorphisms*, preprint 2013. [1](#)
- [16] Harald Lindner, *A remark on Mackey functors*, Manuscripta Mathematica **18** (1976) 273–278.
- [17] Saunders Mac Lane, *Natural associativity and commutativity*, Rice University Studies **49** (1963) 28–46.
- [18] Bodo Pareigis, *A non-commutative non-cocommutative Hopf algebra in nature*, Journal of Algebra **70** (1981) 356–374.
- [19] Ross Street, *Quantum Groups: A Path to Current Algebra*, Australian Mathematical Society Lecture Series **19** (Cambridge University Press, 2007).
- [20] D. Tambara, *The coendomorphism bialgebra of an algebra*, Journal of The Faculty of Science, The University of Tokyo, Section IA, Mathematics, **37** (1990) 425–456.
- [21] K.-H. Ulbrich, *On Hopf algebras and rigid monoidal categories*, Israel Journal of Math. **72** (1990) 252–256.

DEPARTMENT OF MATHEMATICS, MACQUARIE UNIVERSITY, NSW 2109, AUSTRALIA  
*E-mail address*: `steve.lack@mq.edu.au`

DEPARTMENT OF MATHEMATICS, MACQUARIE UNIVERSITY, NSW 2109, AUSTRALIA  
*E-mail address*: `ross.street@mq.edu.au`